

Apple Inc. Revenue Forecasting

Time Series Regression — Portfolio Project Scenario

Scenario

You are a junior analyst at a small equity research team. Your manager has pulled historical quarterly revenue data for Apple Inc. (AAPL) from a standardized financial dataset and has asked you to answer a simple but important question: can Apple's revenue be reasonably predicted as a function of time alone, without relying on external economic indicators or company guidance?

Your manager has been explicit about one constraint: keep the model simple and defensible. A linear trend fit to historical data is the starting point. If the model performs reasonably on the test set, it becomes a baseline. If it does not, understanding why it fails for reasons such as seasonality, nonlinear growth, structural breaks are also useful to know.

Your Task

Build a time series regression model using only historical quarterly revenue figures and a sequential time index. Specifically, you are asked to:

1. Prepare the data by isolating Apple's quarterly revenue and creating a numeric time variable to represent each sequential quarter.
2. Split the data into a training set and a held out test set, reserving the most recent quarters to evaluate your model against real, known outcomes.
3. Fit an OLS regression model using the training data and generate revenue forecasts for the held out period.
4. Report your forecasts as a range using an 80% confidence interval rather than a single point estimate, since your manager wants to communicate uncertainty honestly rather than with false precision.

Why It Matters

This project demonstrates the core workflow of a basic time series forecast: preparing time as a numeric predictor, separating training and testing data to avoid overfitting, fitting a regression model, and communicating a forecast with a range instead of false precision. Using real Apple financial data ties this directly to the time series forecasting component of the portfolio